

Advanced Certification in Agentic and Generative AI

Course Overview – First Lecture
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Welcome to the GenAI C2 Program



Program Snapshot

- **Duration:** 27 weeks (January–July 2026)
- **Total Hours:** 105 hours (60 lecture + 30 project + 9 assessment + 6 campus)
- **Modules:** 10 progressive modules covering the full GenAI stack
- **Capstone:** 40+ hours building a production agentic system

This first session provides a roadmap of the entire program so you know exactly what to expect.



What You Will Learn Today

Session Learning Objectives

1. **Understand** the overall program structure and 10-module progression
2. **Identify** the balance between foundational knowledge and hands-on application
3. **Recognize** the tools, frameworks, and systems you will master
4. **Appreciate** the learning pathway from understanding LLMs to deploying agents
5. **Locate** key resources, schedules, and support channels



The Big Picture

This program follows a **progressive skill-building** approach. Each module builds on the previous, culminating in a production-grade capstone project.

Progressive Learning Architecture

- **Phase 1 – Understanding** (Weeks 1–4): GenAI foundations and prompt engineering
- **Phase 2 – Architecture** (Weeks 5–6): Transformer internals and LLM APIs
- **Phase 3 – Retrieval & Agents** (Weeks 7–12): RAG, agentic systems, scientific ML
- **Phase 4 – Model Building** (Weeks 13–16): Fine-tuning and pre-training custom models
- **Phase 5 – Production** (Weeks 17–27): LLM Ops, deployment, and capstone

You will go from *using* LLMs to *building, deploying, and evaluating* them.



Module Overview

Modules 1–5

- **M1:** GenAI Foundations (6h)
- **M2:** Prompt Engineering (6h)
- **M3:** LLM Architecture & APIs (6h)
- **M4:** RAG & Retrieval Systems (6h)
- **M5:** Agentic AI Foundations (6h)

Modules 6–10

- **M6:** Scientific ML & PINNs (6h)
- **M7:** Fine-Tuning & RLHF (6h)
- **M8:** Pre-Training Models (6h)
- **M9:** LLMOps & Production (9h)
- **M10:** Capstone Project (40h+)

Key Design Principle

Every module includes **theory, coding labs, and a mini-project**. You learn by building.



Time Allocation Across the Program

The 105 program hours are distributed to maximize hands-on learning:

Lecture Hours (60h)

- GenAI Foundations: 12h (M1–M2)
- Architecture & Retrieval: 12h (M3–M4)
- Agents & Scientific ML: 12h (M5–M6)
- Model Building: 12h (M7–M8)
- Production & Capstone: 12h (M9–M10)

Beyond Lectures (45h)

- Mini-Projects: 20h (8 projects)
- Tutorials with TAs: 10h
- Quizzes & Exams: 9h
- Campus Visits: 6h

Capstone Extra

40+ hours of guided project work on top of lectures.



What is Generative AI?

Definition

- **Generative AI** refers to AI systems that can create new content—text, images, code, audio—by learning patterns from training data
- Unlike **discriminative models** (which classify or predict), generative models **produce** novel outputs
- Modern GenAI is powered by **Large Language Models** (LLMs) built on the Transformer architecture

The field has seen explosive growth since 2022. Today, organizations across every industry are deploying GenAI solutions for code generation, customer support, content creation, data analysis, and autonomous decision-making through **agentic systems**.



The LLM Development Stack

Five-Level Framework for GenAI

1. **Foundation Models:** Pre-trained LLMs (GPT, Llama, Mistral, TinyLlama)
2. **Model Adaptation:** Prompt engineering, RAG, fine-tuning
3. **Applications:** Chatbots, assistants, reasoning agents
4. **Integration:** AI embedded in products and workflows
5. **Innovation:** Novel applications, multi-agent systems, autonomous agents

This course takes you through **all five levels** progressively.



Foundations: Understanding LLMs (M1--M3)

M1: GenAI Foundations

- Organizational AI strategy
- GenAI use cases by industry
- What are LLMs and how they work
- Development environment setup

M2: Prompt Engineering

- Zero-shot, one-shot, few-shot
- Chain-of-Thought reasoning
- Role-based and instruction prompting
- Prompt optimization strategies

M3: LLM Architecture & APIs

- Transformer architecture deep-dive
- Attention mechanisms (self, multi-head)
- Tokenization (BPE, WordPiece)
- API parameters: temperature, top-p, top-k
- Inference optimization

Why This Matters

Understanding how LLMs work internally is **essential** for building reliable agents and optimizing production systems.



Retrieval & Agents (M4--M5)

M4: RAG & Retrieval Systems

- What is Retrieval-Augmented Generation
- Vector databases (Pinecone, Weaviate, Qdrant)
- Semantic search and embeddings
- Hallucination mitigation
- RAG evaluation metrics

M5: Agentic AI Foundations

- Agent anatomy: perception, decision, action
- ReAct framework (reasoning + acting)
- Tool use and function calling
- Planning paradigms
- Model Context Protocol (MCP)

Hands-On Deliverables

Mini-Project 4: Build a RAG Q&A system tools

Mini-Project 5: Build a reasoning agent with

Scientific ML & Physics-Informed AI (M6)



Module 6 Coverage

- What is Scientific Machine Learning
- Physics-Informed Neural Networks (PINNs)
- Domain knowledge integration and constraint incorporation
- Neurosymbolic systems and hybrid approaches

Unique to This Program

This module brings together **scientific computing** and **deep learning**—a growing area where AI models respect physical laws and conservation principles. This bridges the gap between traditional numerical methods and modern neural network approaches.



Model Building: Fine-Tuning & Pre-Training (M7--M8)

M7: Fine-Tuning & RLHF

- Supervised fine-tuning (SFT)
- LoRA and QLoRA techniques
- Instruction tuning
- RLHF and DPO alignment
- Domain-specific adaptation

M8: Pre-Training Custom Models

- Causal and masked language modeling
- Data collection and preparation
- Tokenizer training
- Scaling laws and training infrastructure
- Building $<1\text{B}$ parameter models

What You Will Build

Mini-Project 7: Fine-tune a domain model

Mini-Project 8: Pre-train a small language

Production Agentic Systems (M9)



Module 9: LLMOps & Production (9 hours – Expanded)

- **Lecture 1 – Infrastructure (3h):** Docker, quantization, serving frameworks, FastAPI
- **Lecture 2 – Agent Orchestration (3h):** LangGraph workflows, multi-agent coordination, tool registries, agent state management
- **Lecture 3 – Safety & Monitoring (3h):** Guardrails, cost tracking, Prometheus/Grafana monitoring, CI/CD pipelines

This is the **largest single module** at 9 hours, reflecting the importance of production readiness.



Capstone: Your Production GenAI System (M10)

Capstone Requirements (40+ Hours)

- **Component 1:** Custom model (pre-trained or fine-tuned)
- **Component 2:** Application layer (agentic system or RAG pipeline)
- **Component 3:** Production deployment (containerized, API endpoints)
- **Component 4:** Safety guardrails and monitoring
- **Component 5:** Comprehensive evaluation and documentation

Choose Your Theme

30 capstone themes available—from healthcare agents to code-generation assistants to scientific computing pipelines. You choose based on your interests.



Frameworks & Tools You Will Master

AI & Agent Frameworks

- LangChain: agent creation, tool use
- LangGraph: complex agentic workflows
- ReAct: reasoning-acting loops
- Model Context Protocol (MCP)
- Hugging Face Transformers

Infrastructure & APIs

- OpenAI, Anthropic, Google APIs
- Pinecone, Weaviate vector databases
- Docker and containerization
- FastAPI for serving models
- Prometheus and Grafana for monitoring

Implementation Focus

Every framework listed above will be **implemented by you** in labs and projects—not just discussed in lecture.



Frameworks & Tools (continued)

Model Training Stack

- PyTorch for model development
- PEFT (LoRA, QLoRA) for fine-tuning
- Weights & Biases for experiment tracking
- DeepSpeed for distributed training

Development Tools

- Python 3.10+ environment
- Jupyter notebooks for exploration
- Git/GitHub for version control
- VS Code or your preferred IDE

We will help you set up the complete development environment in Module 1. No prior ML tooling experience is assumed.



GenAI in Practice: Working with LLM APIs

What You Will Do

- Call production APIs from OpenAI, Anthropic, and Google
- Control generation with temperature, top-p, top-k, and penalty parameters
- Manage token budgets and context windows effectively
- Build multi-turn conversational applications

Industry Context

Companies like OpenAI (GPT-4), Anthropic (Claude), Google (Gemini), and Meta (Llama) provide the models you will learn to use, fine-tune, and deploy. Understanding how to work with these APIs is the **entry point** to production GenAI.



GenAI in Practice: Building Agentic Systems

Agentic AI – The Frontier

- Agents combine LLM reasoning with **tool use** and **planning**
- The ReAct pattern: Thought → Action → Observation loop
- Multi-agent systems coordinate specialized agents for complex tasks
- Production agents require safety guardrails, monitoring, and cost controls
- This is where the industry is heading—and where this course focuses

Agentic systems represent the next evolution of GenAI: AI that does not just *respond* but actively *reasons*, *plans*, and *acts* to accomplish goals.



GenAI in Practice: RAG & Vector Databases

Retrieval-Augmented Generation

- RAG grounds LLM outputs in **real, retrievable data**—reducing hallucinations
- Documents are embedded as vectors and stored in specialized databases
- At query time, relevant documents are retrieved and injected into the LLM context
- Systems like Pinecone, Weaviate, and Qdrant power production RAG pipelines
- RAG is a core component of most production agentic systems

You will build a complete RAG pipeline from scratch in Module 4.



GenAI in Practice: Custom Model Development

Building Your Own Models

- Fine-tune existing models with LoRA/QLoRA for domain-specific tasks
- Pre-train small language models (<1B parameters) from scratch
- Align models with human preferences using RLHF and DPO
- Evaluate model quality with established benchmarks
- Deploy optimized models using quantization and distillation

Understanding how models are built gives you a **fundamental advantage** when debugging, optimizing, and deploying AI systems in production.



Mini-Projects: What You Will Build

Foundation Projects

- **MP1:** Explore GenAI use cases
- **MP2:** Prompt optimization lab
- **MP3:** API parameter tuning
- **MP4:** RAG Q&A system

Advanced Projects

- **MP5:** Multi-step reasoning agent
- **MP6:** Scientific ML application
- **MP7:** Fine-tune a domain model
- **MP8:** Pre-train a small LM

Portfolio Building

Each mini-project produces a **GitHub repository** with documentation. By course end, you will have 10+ repositories demonstrating production-level work.



Learning Paths Within the Program

Choose Your Focus Area

- **Path 1 – API-First Developer** (14 weeks): M1 → M2 → M3 → M5 → M9 → M10
- **Path 2 – RAG Specialist** (16 weeks): Adds M4 for retrieval expertise
- **Path 3 – Agentic Systems Builder** (16 weeks): Deep focus on M5, M9, M10
- **Path 4 – Model Developer** (18 weeks): Adds M7, M8 for training expertise
- **Path 5 – Full Stack AI Developer** (27 weeks): All 10 modules, 396 concepts

All paths share the same foundation (M1–M3). Your capstone theme determines your specialization.



Skills You Will Develop

Technical Skills

- LLM architecture understanding
- Prompt engineering mastery
- RAG pipeline development
- Agent design and orchestration
- Model fine-tuning and training

Production Skills

- Containerized ML deployment
- API design for AI systems
- Monitoring and observability
- Safety guardrails implementation
- Cost optimization strategies

Career Relevance

These skills map directly to roles such as **AI Engineer**, **ML Engineer**, **Data Scientist**, **Applied Scientist**, and **GenAI Product Manager**.



Your Portfolio at Completion

Concrete Deliverables

- Fine-tuned <1B parameter model
- Complete RAG Q&A system
- Multi-step reasoning agent
- Production agentic application
- Capstone: full deployed system

Professional Artifacts

- 10+ GitHub repositories
- Complete documentation per project
- Deployment guides
- Model cards and evaluation reports
- Technical presentation skills

Career-Ready

Your portfolio demonstrates both **model expertise** and **production deployment skills**—exactly what hiring teams look for.



Weekly Schedule Overview

Weeks	Module	Hours	Focus
1-2	M1: GenAI Foundations	6	Understanding
3-4	M2: Prompt Engineering	6	Prompting
5-6	M3: LLM Architecture	6	Architecture
7-8	M4: RAG Systems	6	Retrieval
9-10	M5: Agentic AI	6	Agents
11-12	M6: Scientific ML	6	Domain AI

Detailed weekly schedules will be posted in advance at the start of each module.



Weekly Schedule Overview (continued)

Weeks	Module	Hours	Focus
13-14	M7: Fine-Tuning & RLHF	6	Model Adaptation
15-16	M8: Pre-Training	6	Model Building
17-19	M9: LLMOps & Production	9	Deployment
20-27	M10: Capstone	40+	Full System

Each lecture is 3 hours long. There are 20 lectures total across the program, with tutorials and project work supplementing each module.



Assessment Structure

Formative Assessments

- Module quizzes after each module
- Mini-project submissions (8 total)
- Coding lab participation
- Peer review exercises

Capstone Evaluation

- System testing and benchmarking
- Technical presentation and demo
- Q&A defense session
- Documentation and model card review

Grading Philosophy

Assessment focuses on **demonstrated competence**—can you build, deploy, and explain a working system? Theory knowledge supports but does not replace practical skills.

Resources & Support



Learning Resources

- Lecture slides and recordings
- Code repositories for each module
- Recommended readings and papers
- Tutorial notebooks with solutions

Support Channels

- Office hours with Prof. Ganesan
- TA-led tutorial sessions
- Course discussion forum
- Peer study groups

Contact Information

Instructor: Prof. Sashikumaar Ganesan | **Email:** sashi@iisc.ac.in | **Office Hours:** To be announced



Summary & Key Takeaways

What to Remember from Today

- This is a **27-week progressive program** covering the full GenAI development stack
- You will learn to **use, build, fine-tune, and deploy** LLMs and agentic systems
- Every module includes **theory, coding labs, and a hands-on project**
- The capstone produces a **production-grade** AI system for your portfolio
- Multiple **learning paths** let you specialize based on your career goals

Next Steps



This Week

- Set up your Python development environment
- Join the course discussion forum
- Review the Module 1 pre-reading materials
- Start thinking about GenAI use cases in your domain

Coming Up Next

- **Lecture 2:** AI-first organizations and LLM fundamentals
- **Lab 1:** Development environment setup and first GenAI prompt
- **Mini-Project 1:** Explore GenAI use cases



Questions & Discussion

Let us take a few minutes for questions before we move on. This is a good time to clarify anything about the program structure, expectations, or logistics.

Open Discussion

- Questions about the program structure or schedule?
- Specific tools or frameworks you want to ensure are covered?
- Any logistical or setup questions?

We are here to make this the best learning experience possible.

Let's Begin

*Next: What is Generative AI?
Module 1, Lecture 1*